

1. Project Title: Covid-19 Detection

2. Team Members:

- Drew Quashie
- Amanda Ogbonna
- Loleyi Oluwatomisin
- Richard Alonso Garcia

3. Faculty Advisor:

Name: Dr. Zahra Nematzadeh

Email: znematzadeh@fit.edu

4. Client:

- Medical Professionals
- Potential Covid Patients

5. Meeting Dates

1. Tuesday August 25th 11:15am
- Olin Engineering Complex room 353

6. Goals and Motivations:

- Allow users to detect COVID-19 early using accessible devices like smartphones or computers.
- Develop a web app that can predict COVID-19 infections by analyzing coughing sounds.
- Provide real-time risk assessments and longitudinal health tracking to assist both patients and medical professionals.

7. Approach:

- **Phase 1: Data Preprocessing & Augmentation:** Extract Mel-Frequency Cepstral Coefficients (MFCCs) and spectrogram images from raw audio signals in the CoughVid dataset (~2,800 recordings). Apply signal normalization, SpecAugment, pitch shifting, time shifting/stretching, and additive noise to improve robustness. Partition data into 70% training, 15% validation, and 15% test sets with k-fold cross-validation.
- **Phase 2: Deep Learning Model Development:** Build, train, and fine-tune Convolutional Neural Networks (CNNs) and leverage transfer learning architectures (e.g., ResNet) using spectrogram inputs to classify COVID-19 status.
- **Phase 3: Model Evaluation:** Measure performance using Accuracy, Precision, Recall, F1 Score, and ROC-AUC metrics on an independent test dataset.
- **Phase 4: Web Application Integration:** Develop a full-stack web application capable of capturing audio in real time, running feature extraction, serving backend inference predictions, and displaying risk feedback alongside guidance on next steps.

8. Novel Features/Functionalities

- **Web-Based Audio Capture:** Browser-accessible interface enabling users to record cough sounds directly without requiring specialized hardware.
- **Automated Risk Assessment & Guidance:** Real-time output providing risk levels (Low, Medium, High) paired with clinical recommendations (e.g., self-isolation, contacting a healthcare provider).
- **Longitudinal Cough Tracking Dashboard:** Interactive health tracking visualization monitoring daily cough frequencies and health trends over time.

9. Algorithms and Tools

Domain	Tool / Technique
Frontend Framework	React, Web Audio API
Backend & API	Python, FastAPI
Machine Learning	PyTorch, Librosa (Feature Extraction)
Model Architectures	Convolutional Neural Networks (CNNs), ResNet (Transfer Learning)
Dataset	CoughVid Dataset (2,800 audio recordings with status and metadata)
Evaluation Metrics	Accuracy, Precision, Recall, F1 Score, ROC-AUC

10. Technical Challenges

- **Audio Noise Variability:** High ambient background noise and microphone hardware discrepancies across user web browsers require robust audio preprocessing, filtering, and data augmentation.
- **Real-Time Inference Latency:** Optimizing the pipeline between browser-based web recording, audio feature conversion, and server-side PyTorch execution to ensure near-instant feedback for the user.
- **Dataset Class Imbalance & Generalization:** Addressing potential sample distribution biases in cough audio to maintain model accuracy across diverse demographic profiles and symptom severities.

11. Milestone 1 (September 28th)

- Complete project planning, requirements documentation, and system architecture design.
- Set up CoughVid dataset pipeline: audio filtering, feature extraction (MFCCs/Spectrograms), and normalization.
- Implement data augmentation techniques (additive noise, pitch/time shifting, SpecAugment).
- Build baseline PyTorch CNN model structure and React frontend audio recording component.

12. Milestone 2 (October 26th)

- Train and fine-tune advanced CNN models and transfer learning architectures (e.g., ResNet).
- Complete initial model evaluation and generate performance reports (F1 Score, ROC-AUC).
- Develop FastAPI backend endpoints for handling client audio uploads and running inference pipelines.
- Build initial React web user interface components for displaying prediction results and risk feedback.

13. Milestone 3 (November 23rd)

- Integrate React frontend, FastAPI backend, and PyTorch model for end-to-end real-time predictions.
- Implement the longitudinal cough tracking visualization dashboard.
- Complete full system testing, performance optimization, and bug resolution.
- Finalize all project deliverables: demonstration video, poster, and final project presentation and publish paper.

14. Milestone 1 Task Matrix

Task Description	Assigned Team Member	Status
Requirements & Project Plan Documentation	Amanda Ogbonna	Completed
CoughVid Audio Data Preprocessing & MFCC/Spectrogram Pipeline	Drew Quashie	In Progress
Audio Augmentation Pipeline (Noise, Pitch & Time Shifts)	Drew Quashie	In Progress
Baseline PyTorch CNN Architecture Development	Loleyi Oluwatomisin Amanda Ogbonna	Pending
React Audio Recording Interface Setup	Richard Alonso Garcia	Pending

15. Faculty Approval

Faculty Advisor Name: Dr. Zahra Nematzadeh

Signature: _____

Date: _____